

Fundamental Limitation in Explaining AI

A Quadrilemma among Environment Complexity, AI Performance, Interpretability, and Faithfulness

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Materials available here!

- arXiv: <https://arxiv.org/abs/2605.24727>



- The presentation slides:

https://ash-suzuki.github.io/material/xai_limitation_talk.pdf



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- 2011-2020: Bachelor (Engineering, 2015), Master (Information Science and Technology (IST), 2017), PhD (IST, 2020), The University of Tokyo
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Research interests:

- Methods for AI (differential geometry, information theory, deep learning)
- Theoretical analysis of AIs (statistics, information theory, computability theory)

Outline

1. Motivation
2. Problem Formulation
3. Formulating the 4 conditions
4. Main Result
5. Conclusion
6. References

Success of Als

Success of AIs

- McKinsey estimates its annual economic value to be **\$2.6–\$4.4 trillion** [10].
- Goldman Sachs estimates it could boost **global GDP by about 7% (approximately \$7 trillion)** over the next decade [4].

Social demand in explaining AIs

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Example (Action Plans by NIST (US) [11])

- “Address general risks associated with a lack of **explainability and transparency** in GAI systems” (GV-4.1-001)
- “Implement interpretability and explainability methods to evaluate GAI system decisions and verify alignment with intended purpose.” (MS-4.2-003)

Existing explanations are not completely “faithful”

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- **Requiring scalarization of the output:** LIME [13], SHAP [9], integrated gradients [18], Grad-CAM [16] etc.
- **Generating a text:** LLM-based approaches, e.g., [5–7].
- **surrogate models:** e.g., LIME [13], Anchors [14], GLocalX [17], GLEAMS [19].

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Non faithful explanation example: LIME

Example (LIME (Local Interpretable Model-agnostic Explanations) [13])

- Considers a scalar output and binarization of the original data points, i.e., intentionally **gives up the faithfulness** of the explanation.
- Explanation: local approximation in the binarized space.

Specifically, fix a binarization function $(\bullet)_{\text{bin}} : \mathcal{X} \rightarrow \{0, 1\}^d$ and recover function $(\bullet)_{\text{rec}} : \{0, 1\}^d \rightarrow \mathcal{X}$. The explanation of a function $f : \mathcal{X} \rightarrow \mathbb{R}$ of interest around x is given by $w_x \in \mathbb{R}^{d'}$ that minimizes

$$\sum_{z'_{\text{rec},x} \in \text{neighbor}(x_{\text{bin}})} \text{similarity}(x, z'_{\text{rec},x}) \left(f(z'_{\text{rec},x}) - w_x^\top z'_{\text{rec},x} \right)^2 \quad \text{s.t. } \|w_x\|_0 \leq K.$$

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Before we discuss this problem, we first need to exclude trivial and uninteresting cases.

Exclude trivial cases. 1. When AI's performance is poor

Example (A poor AI case)

Task: AI is asked to answer the user's question.

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Task: AI is asked to answer the user's question.

- **User prompt 1:** Tell me the smallest perfect number?
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- **AI's answer 2:** I am an AI.

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- **User prompt 2:** Where is the capital city of Japan?
- **AI's answer 2:** I am an AI.
- **Explanation:** "The AI just outputs I am an AI., i.e., $AI(x) = \text{I am an AI.}$ "

In this case, while the explanation is faithful, the AI's performance is poor.

Exclude trivial cases. 2. When the task/environment is simple.

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In this case, the AI's performance is perfect, and the explanation is faithful.

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- **AI's answer 2:** The de facto capital city of Japan is Tokyo.
- **Explanation:** (Just gives the source code and the checkpoint)

In this case, while the AI performs well and the explanation is faithful, the explanation is NOT interpretable.

Refined research question

Research question: Can there exist an AI and its explanation such that:

- The AI works in a complicated environment, e.g., natural language Q/A.
- The AI's performance is good.
- The explanation is completely faithful to the AI behavior.
- The explanation is interpretable.

Main answer

Theorem (Informal quadrilemma)

*An AI system and its explanation **CANNOT** simultaneously satisfy all of the following:*

- The AI's operation environment is complicated,*
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Actually, today's talk is mainly about definitions, rather than proofs.

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But... **How** can we mathematically define the **interpretability** of an explanation?

Key trick to avoid the difficulty in formulating the interpretability

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Our idea: Use the **shortness** of the explanation as a necessary condition for the interpretability.

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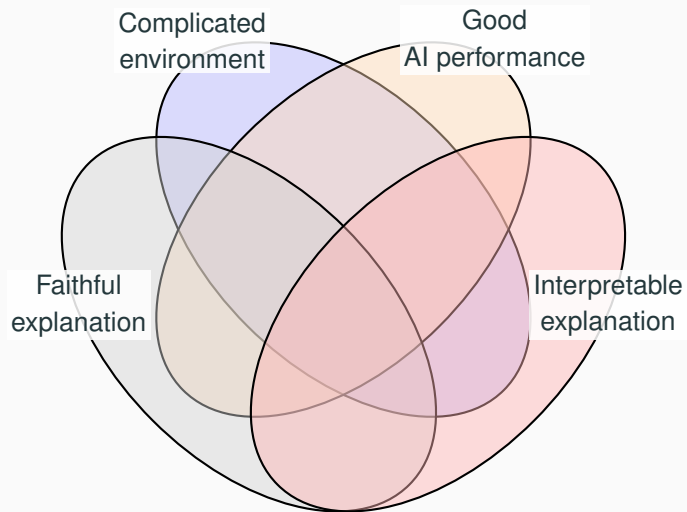
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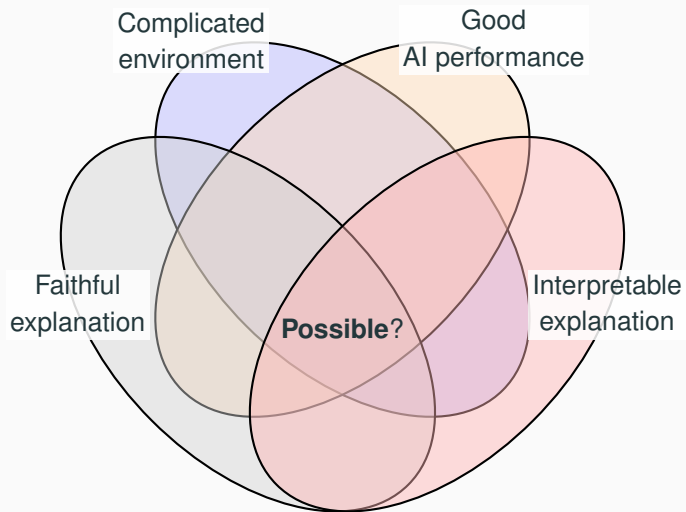
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This is why the pair of the source code and checkpoint (28B bits even for a 7B model with the 4 bit quantization.) of the open model is NOT interpretable.

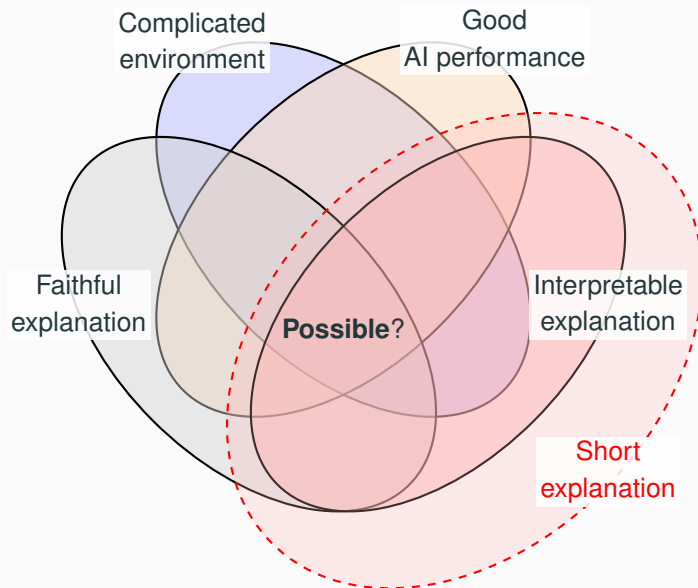
Conceptual picture



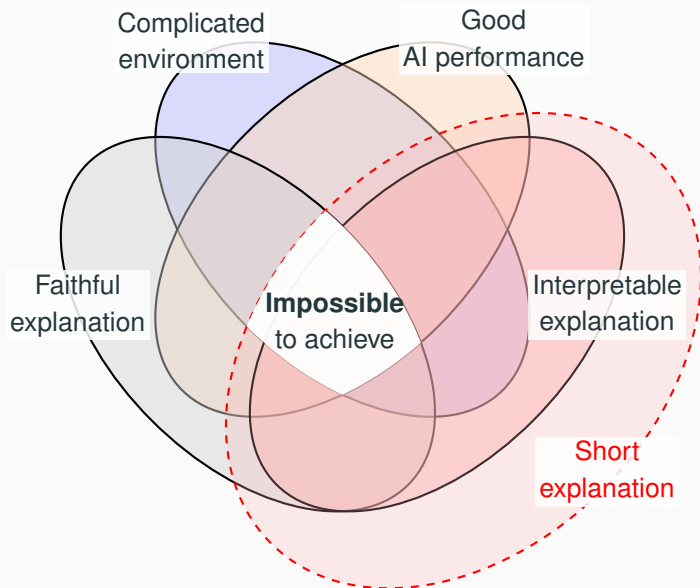
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What we actually prove is...

Theorem (Informal quadrilemma (more formulatable))

*An AI system and its explanation **CANNOT** simultaneously satisfy all of the following:*

- The AI's operation environment is complicated,*
- The AI's performance is good,*
- The explanation is completely faithful to the AI behavior,*
- The explanation is **short**.*

The above statement is much easier to formulate than the original form.

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What do we need to do?

- Formulate an AI itself.
- Formulate the goodness of AI performance.
- Formulate the faithfulness of the explanation.
- Formulate the shortness of the explanation.
- Formulate AI's operation environment

Preliminary: finite-length strings

Definition (Finite-length string)

Let Σ be the letter set that the computer can natively use. Typically $\Sigma = 0, 1$. A **finite-length string** is a finite-length sequence of letters, e.g., 001101011.

The set of all finite-length strings is denoted by $\Sigma^* := \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$.

Note that in a computer, everything needs to be represented as a finite-length string, and the set Σ^* is countably infinite.

Hence, it CANNOT represent the whole of an uncountably infinite set such as the real number set $\mathbb{R}$ or the function class $\{f : \Sigma^* \rightarrow \Sigma^*\}$.

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- **Output** $y = f_p(u, x) \in \Sigma^*$: response to the users, internal steps, etc.

where $f_p : \Sigma^* \rightarrow \Sigma^*$ is the function implemented by p .

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It receives a user prompt x pseudo-random numbers and returns an image y , showing a pseudo-stochastic behavior. The termination random variable set $\mathcal{T}_x$ consists of fixed-length sequences.

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It also receives a user prompt x pseudo-random numbers and returns an image y , showing a pseudo-stochastic behavior. The termination random variable set $\mathcal{T}_x$ (all random number sequences that output EOS in the end) depends on x .

Behavior as a conditional probability mass function

Definition (AI-induced conditional probability mass function)

Let $\text{Prob}(\mathbf{u})$ be a probability mass virtually allocated to pseudo random number sequence $\mathbf{u}$.

For a stochastic input-output AI A with the termination condition given by $\mathcal{T}_x$, define Q_A by

$$Q_A(y \mid x) = \sum_{\mathbf{u} \in \mathcal{T}_x} \text{Prob}(\mathbf{u}) \mathbb{1}(f(\mathbf{u}, x) = y).$$

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The goal of AI explanation: **to explain** Q_A .

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Formulation of AI Performance: perplexity

As a metric of the AI performance, we simply use **perplexity**, widely used in NLP (e.g., [12]).

Definition (Perplexity)

Given an input $x \in \mathcal{X}$ and an output $y \in \mathcal{Y}$, the **perplexity** of a conditional probability mass function Q is

$$\frac{1}{Q(y \mid x)}.$$

Given a true distribution P on $\mathcal{X} \times \mathcal{Y}$, the expected logarithmic perplexity is

$$\mathbb{E}_{X,Y \sim P} \left[-\log Q(Y \mid X) \right].$$

Remark

A Small expected logarithmic perplexity means good AI performance.

Formulation of completely faithful explanations

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Definition (Interpretation function)

Fix a letter set Λ of natural languages (e.g., Unicode). An **interpretation function** is a **computable** function

$$\text{Interpret} : \subseteq \Lambda^* \times \mathbb{N} \times \mathcal{Z} \xrightarrow{\text{comp}} \mathbb{Q}$$

that interprets an explanation string as an arbitrary-precision approximation of a real-valued function.

Notes on a computable function

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In the above, we can replace Python with another programming language, e.g., C, because we can write a C emulator in Python and a Python emulator in C.

A function implemented in a programming language may not define the return value for some inputs owing to an infinite loop. Such a function is called a partial function, and is denoted like $f : \subseteq \mathcal{X} \rightarrow \mathcal{Y}$, instead of $f : \mathcal{X} \rightarrow \mathcal{Y}$.

When $f(x)$ is defined, we write $f(x) \downarrow$.

A formal definition of computable functions (Preliminary)

In the following, we define the set of string-input/output computable function.

We can convert the definition of computability to define the computability of functions with general source set $\mathcal{X}$ and target set $\mathcal{Y}$ by a fixed encoding rule, i.e., maps $\mathcal{X} \rightarrow \Sigma^*$ and $\mathcal{Y} \rightarrow \Sigma^*$, as long as $\mathcal{X}, \mathcal{Y}$ are at most countable.

A formal definition of computable functions (Part 1: primitive recursive)

Definition (Primitive recursive functions (string version))

Let the alphabet set Σ be a finite set such that $|\Sigma| \geq 2$ and let Σ^* denote the set of finite-length strings, i.e., $\Sigma^* := \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$. We define the set of primitive recursive functions according to the following rules.

- $\text{emp}, \text{append}_a : \Sigma^* \rightarrow \Sigma^*$ (for all $a \in \Sigma$) are primitive recursive functions, where
 - $\text{emp}(x) = ()$.
 - $\text{append}_a(x) = x \cdot a$ for all $a \in \Sigma$.
- If $f, g : \Sigma^* \rightarrow \Sigma^*$ are primitive recursive function, then $\text{comp}[f, g] : \Sigma^* \rightarrow \Sigma^*$ is also a primitive recursive function, where
 - $\text{comp}[f, g](x) = g(f(x))$.

A formal definition of computable functions (Part 1: primitive recursive)

Definition (Primitive recursive functions (string version) (continued))

Let the alphabet set Σ be a finite set such that $|\Sigma| \geq 2$ and let Σ^* denote the set of finite-length strings, i.e., $\Sigma^* := \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \dots$. We define the set of primitive recursive functions according to the following rules.

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 - $\text{iter}[c, (t_a)_{a \in \Sigma}](x) = s_{|c|}$ where
 - $c = c(x)$
 - $s_k := \begin{cases} x & \text{if } k = 0, \\ t_{c_{k-1}}(s_{k-1}) & \text{if } k = 1, 2, \dots, |c|. \end{cases}$

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Let $\mathcal{X}$ and $\mathcal{Y}$ be sets. If $f(x) \in \mathcal{Y}$ is defined for a $x \in \mathcal{X}$, then we write $f(x) \downarrow$.

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We say that f is a **partial function** from the source set $\mathcal{X}$ to the target set $\mathcal{Y}$, and write $f : \subseteq \mathcal{X} \rightarrow \mathcal{Y}$.

If $f(x)$ is not defined for a $x \in \mathcal{X}$, then we write $f(x) \uparrow$.

- In the computability theory context, $f(x) \uparrow$ indicates the program realizing f falls into an infinite loop.

A formal definition of computable functions (Part 2: partially computable)

Definition ((Partially) computable functions (string version))

We define the set of (partially) computable functions according to the following rules.

- $\text{emp}, \text{append}_a : \Sigma^* \rightarrow \Sigma^*$ are partially computable functions.
- If $f, g : \subseteq \Sigma^* \rightarrow \Sigma^*$ are partially computable function, then $\text{comp}[f, g] : \Sigma^* \rightarrow \Sigma^*$ is also partially computable, where
 - $\text{comp}[f, g](x) = g(f(x))$ if

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Definition ((Partially) computable functions (string version) (continued))

- $(\text{emp}, \text{append}_a, \text{comp}[f, g])$ were partially computable)
- If $c, t_a : \subseteq \Sigma^* \rightarrow \Sigma^*$ are partially computable for all $a \in \Sigma$, then $\text{iter}[c, (t_a)_{a \in \Sigma}]$ is also partially computable, where
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if $t_{c_{k-1}}(s_{k-1})$ is defined for $k = 1, 2, \dots, |c|$; otherwise undefined.

A formal definition of computable functions (Part 2: partially computable)

Definition ((Partially) computable functions (string version) (continued))

- $(\text{emp}, \text{append}_a, \text{comp}[f, g], \text{iter}[c, (t_a)_{a \in \Sigma}])$ were partially computable)
- If $p, t : \subseteq \Sigma^* \rightarrow \Sigma^*$ are partially computable for all $a \in \Sigma$, then $\text{while}[p, t]$ is also partially computable, where
- $\text{while}[p, t](x) = s_{k^*}$, where
 - $s_k := \begin{cases} x & \text{if } k = 0, \\ t(s_{k-1}) & \text{if } k > 0, \end{cases}$
 - $k^* := \min\{k \mid p(s_{k'}) \downarrow \text{ for all } k' = 0, 1, \dots, k \text{ and } p(s_k) = ()\}$.
if

A formal definition of computable functions (Part 2: partially computable)

Definition ((Partially) computable functions (string version) (continued))

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 - $k^* := \min\{k \mid p(s_{k'}) \downarrow \text{ for all } k' = 0, 1, \dots, k \text{ and } p(s_k) = ()\}$.
if $\min\{k \mid p(s_{k'}) \downarrow \text{ for all } k' = 0, 1, \dots, k \text{ and } p(s_k) = ()\}$ is non-empty; otherwise undefined.

An infinite loop caused by "while"

The following pseudocode of $\text{while}[p, t]$ shows why an infinite loop may occur.

- **Input:** x .
- **Initialize** $s_0 := x$.
- **For** $k = 0, 1, 2, \dots$:
- **Compute** $s_k := t(s_{k-1})$.
- **If** $p(s_k) = ()$, then **return** s_k ; **otherwise continue**.

Possibly, $p(s_k) = ()$ is never satisfied; then it leads to an infinite loop.

Faithfulness as reproducibility under a fixed interpretation rule

Definition (Completely faithful explanation)

A string $e \in \Lambda^*$ is a completely faithful explanation of a function

$$f : \mathcal{Z} \xrightarrow{\text{comp}} \mathbb{R}$$

under a fixed computable function $\text{Interpret} : \Lambda^* \times \mathbb{N} \times \mathcal{Z} \rightarrow \mathbb{Q}$ if, for every $z \in \mathcal{Z}$ and every $k \in \mathbb{N}$,

$$|\text{Interpret}(e, k, z) - f(z)| < |\Sigma|^{-k}.$$

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$$|\text{Interpret}(e, k, z) - f(z)| < |\Sigma|^{-k}.$$

Remark

Considering the approximation is necessary since a computer cannot directly handle irrational numbers.

Formulating the length of an explanation

The length of the explanation is $|e|$, the length as a sequence.

$$|A \text{ good } AI| = 9.$$

Key idea for formulating the operating environment complexity

How do we quantify the complexity of the input-output relation?

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If the relation is simple, we can find a simple rule that generates the output from the given input.

Our idea: Use **the shortest program length** generating the output from the given input as a measure of the input-output relation complexity.

This idea leads us to the definition of **conditional plain Kolmogorov complexity**.

Formulation of the operating environment complexity

Definition (Conditional plain Kolmogorov complexity)

Fix a **universal conditional function** U (like a Python interpreter). For $x, y \in \Sigma^*$, the **conditional plain Kolmogorov complexity** is defined by

$$C_U(y \mid x) := \min \{ |p| \mid U(\bar{x} \cdot p) \downarrow = y \}.$$

$\bar{\cdot}$ is a fixed self-delimiting encoder function.

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We use $\mathbb{E}_{X,Y} C_U(Y \mid X)$ as a metric of the operating environment complexity.

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Note: changing U only changes an additive constant independent of x and y .

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1. Motivation

2. Problem Formulation

3. Formulating the 4 conditions

4. Main Result

5. Conclusion

6. References

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Fundamental Quadrilemma: a rough formula

Our theorem roughly states that:

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Implication: If the operating environment (task) is complex, then the RHS is large.

Then, either of the following holds:

- The perplexity is large (poor performance), or
- The (faithful) explanation is long (uninterpretable explanation).

Fundamental Quadrilemma: mathematical form

Theorem (Fundamental Quadrilemma)

Fix a universal function U , an encoding function an alphabet Σ , an interpretation function with character set Λ , and $L_\Lambda := \max_{\lambda \in \Lambda} |\text{Enc}_\Lambda^{\text{PF}}(\lambda)|$.

Suppose that for every $x \in \mathcal{X}$, a completely faithful explanation e_x of $Q_A(\cdot | x)$ is given. For every probability distribution P on $\mathcal{X} \times \mathcal{Y}$,

$$\begin{aligned} \mathbb{E}_{X,Y \sim P} [-\log Q_A(Y | X)] + \mathbb{E}_{X,Y \sim P} [L_\Lambda |e_X| + 2 \log (L_\Lambda |e_X| + 1)] \\ \geq \mathbb{E}_{X,Y \sim P} C_U(Y | X) + c. \end{aligned}$$

The constant c does not depend on A , x , y , or P .

If Λ is Unicode characters and UTF-8 for encoding, $L_\Lambda = 32$.

Meaning of the inequality

Remark (Trade-off)

The theorem implies that the following cannot all be small or favorable:

1. performance (expected logarithmic perplexity): $\mathbb{E}[-\log Q_A(Y | X)]$,
2. expected faithful explanation length, $\mathbb{E}[|e_X|]$,
3. environment complexity, $\mathbb{E}C_U(Y | X)$.

Remark

Recall that a lengthy explanation is NOT interpretable.

Implication for AI governance

Remark (Governance implication)

In many practical applications:

1. the operation environment is complex,
2. good AI performance is required,
3. interpretable explanations are required.

Therefore, the condition to give up is usually complete faithfulness.

Remark

AI governance should be designed on the premise that AI explanations are incomplete in faithfulness.

Local and global explanations

Remark

The theorem is stated for local explanations e_x of $Q_A(\cdot \mid x)$. A global explanation of $Q_A(\cdot \mid \cdot)$ must be able to recover each local behavior $Q_A(\cdot \mid x)$. Hence, up to fixed constants, the lower-bound logic also applies to global explanations.

Hypothesis on the environment complexity.

Since the environment itself is complex, computing its complexity (e.g., $\mathbb{E}C_U(Y \mid X)$) is, in principle, a difficult problem.

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Nevertheless, our quadrilemma theorem is suggestive. If an AI that achieves the true input-output relation and is written by a short program were realized, then $\mathbb{E}_{X,Y \sim P}C_U(Y \mid X)$ would be small, and the quadrilemma would not be a problem. However, considering the history in which text generation could not achieve sufficient performance until the advent of large-scale language models, one can formulate the hypothesis that such a situation will not occur.

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Summary

Since shortness is a necessary condition for human interpretability, the theorem gives a mathematical quadrilemma among:

1. environment complexity,
2. AI performance,
3. interpretability,
4. complete faithfulness.

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Stochastic input-output AI

Let $\mathcal{X}$ be the input space and $\mathcal{Y}$ be the output space.

Definition (Stochastic input-output AI)

A stochastic input-output AI with domain $\mathcal{X}' \subseteq \mathcal{X}$ is a pair

$$A = (f, \tau),$$

where

$$f : \subseteq (\mathbb{N}_{<n})^* \times \mathcal{X} \xrightarrow{\text{comp}} \mathcal{Y}$$

is the main function, and

$$\tau : \subseteq (\mathbb{N}_{<n})^* \times \mathcal{X} \xrightarrow{\text{comp}} \{0, 1\}$$

is the random-sequence acceptance decision function.

Accepted random sequences

For each input $x \in \mathcal{X}'$, define

$$\mathcal{T}_x := \{\mathbf{u} \in (\mathbb{N}_{<n})^* \mid \tau(\mathbf{u}, x) \downarrow = 1\}.$$

Definition (Key requirements)

For each $x \in \mathcal{X}'$:

1. $\mathcal{T}_x$ is computable,
2. $\mathcal{T}_x$ is prefix-free,
3. a computable stopping bound $N_{\text{stop}}(x, k)$ satisfies

$$1 - \sum_{\substack{\mathbf{u} \in \mathcal{T}_x \\ |\mathbf{u}| \leq N_{\text{stop}}(x, k)}} n^{-|\mathbf{u}|} \leq |\Sigma|^{-k},$$

4. if $\mathbf{u} \in \mathcal{T}_x$, then $(\mathbf{u}, x) \in \text{dom } f$.

Intended stochastic operation

Definition (Intended operation)

Given $x \in \mathcal{X}'$, the AI repeatedly samples symbols from a uniform oracle $\text{Uniform}_{<n}$ and concatenates them into a random string u .

When $u \in \mathcal{T}_x$, it outputs

$$f(u, x)$$

and terminates.

Remark

This framework includes deterministic functions, diffusion models with fixed random-number consumption, and LLMs whose number of consumed random numbers depends on generated tokens and termination conditions.

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Naive idea

A natural first attempt is to measure environment complexity by the Kolmogorov complexity of the true conditional probability mass function:

$$C_U(P(\cdot \mid \cdot)).$$

Proposition (Naive exact-minimization result)

If Q has a completely faithful explanation e with

$$L_\Lambda|e| < C_U(P) - c_{\text{Interpret}},$$

then Q cannot exactly minimize expected logarithmic conditional perplexity.

Remark

This only rules out exact minimization.

Failure of the naive method

Proposition (Failure of the naive lower bound)

Let $\mathcal{X}$ and $\mathcal{Y}$ be countably infinite. For every $n \in \mathbb{N}$ and every $\epsilon > 0$, there exist π , computable P, Q , and a completely faithful explanation e of Q such that

$$\mathbb{E}_{X \sim \pi, Y | X \sim P}[-\log Q(Y | X)] - \min_{Q^*} \mathbb{E}_{X \sim \pi, Y | X \sim P}[-\log Q^*(Y | X)] < \epsilon,$$

but

$$|e| < C_U(P) - n.$$

Remark

Thus $C_U(P)$ does not give the desired performance-explanation trade-off near the optimum.

Why expected conditional Kolmogorov complexity is used

Remark (Chosen complexity indicator)

The theorem uses

$$\mathbb{E}_{X,Y \sim P} C_U(Y \mid X)$$

because it measures the algorithmic complexity of individual true outputs conditioned on inputs.

Remark

Replacing it by entropy alone loses the implication for explanation length, because Gibbs' inequality already gives

$$\mathbb{E}_{X,Y \sim P} [-\log Q(Y \mid X)] \geq H_P(Y \mid X).$$

Gap from entropy

Proposition (Kolmogorov complexity can exceed entropy)

For every $n \in \mathbb{N}$, there exists a probability distribution P such that

$$\mathbb{E}_{X,Y \sim P} K_V(Y \mid X) \geq H_P(Y \mid X) + n.$$

Remark

This shows that the main theorem can yield a stronger lower bound than an entropy-only argument.